less information about determinants of programme participation than we do. (Recall the summary in Table 1.) We seek to learn how the quality of the data used to estimate the probability of programme participation influences the effectiveness of the estimators. We use the same randomized control data and eligible nonparticipant data (ENP) as above, but vary the predictors in $P(X)$. Inclusion of additional predictors is not guaranteed to improve the effectiveness of a matching estimator. As an extreme example of this point, introducing conditioning variables that perfectly classified applicants and nonapplicants would make matching impossible, because there would be no overlap in the support of X given $D=1$ and X given $D=0$, and (A-2) would not be satisfied for any X. This example emphasizes the important point that matching requires X variables that are good enough to obtain conditional independence between Y_0 and D, but that are not "too good" i.e. that predict D perfectly.

Our evidence indicates that matching estimators perform best when variables describing recent unemployment and earnings histories, shown to be important determinants of programme participation in Heckman and Smith (1994), are used to predict participation. All of the estimates reported in Tables 5(a)–5(b) are based on participation models that use this information. Bias is substantially greater if only background demographic variables are used in estimating the probability of participation. Access to information on earnings in the previous year sometimes improves the performance of the matching estimator but the estimator performs best when data on recent labour force participation patterns are available.

We estimate the probability of programme participation, $P(X)$, using four different sets of variables that differ in the type of data used to describe recent labour force dynamics. The names ascribed to the different coarser conditioning sets are "regular", coarse I, coarse II, coarse III. The regressors included in each model and their relationship to the regular P scores are described in the footnotes to Table 6(a). Briefly, coarse I contains only limited demographic information. Coarse models II and III supplement the background data using different information about recent labour force histories and earnings prior to the date of enrollment into the programme. The "regular" models for $P(X)$ participation use information on demographics, earnings and labour force histories.[29]

Table 6(a) shows the importance of building a good model of programme participation. (The final three columns of this table are discussed in the following sections.) Results are reported for the regression-adjusted local linear regression estimator using the rich or regular model for $P(X)$ and the coarser scores. For adult males, the choice of the matching variables X matters greatly. Failure to control for earnings or employment histories (Coarse I) produces a badly biased estimator. Controlling for previous annual earnings alone results in less bias (Coarse II) but augmenting the conditioning set to include information on recent labour force transitions results in a substantial reduction in bias (Coarse III). The safest conclusion to draw from this evidence is that matching on $P(X)$ works well if it is based on a good model of programme participation. For each demographic group, the matching estimator based on our local linear regression model using the best predicting model of programme participation performs as well or better than any other model in the table. For adult women, the use of the richer models for $P(X)$ leads to a slight increase in the bias. For male youth, the bias is substantially higher under the coarse

29. In figures available upon request, we plot the P scores for comparison and control group participants for the different demographic groups. The nonoverlapping support problem is present for all the coarse scores. Use of coarse I exacerbates the support problem, shrinking the overlapping support region, but for other coarse scores there is no particular pattern across the groups.

TABLE 6(a)

Bias from local linear regression matching estimator†
Under alternative predictor models for the probability of programme participation

Quarter	Regular††	Coarse I‡	Coarse II‡	Coarse III‡	SIPP§	Site mismatch§§	No-show§§§
				Adult males			
$t=1$	39 (60)	−390 (51)	−228 (67)	−84 (77)	249 (77)	−184 (110)	58 (38)
$t=2$	39 (64)	−312 (58)	−193 (61)	−39 (88)	123 (79)	−154 (120)	37 (39)
$t=3$	21 (80)	−286 (62)	−153 (57)	−36 (96)	76 (81)	−147 (127)	27 (42)
$t=4$	65 (82)	−231 (63)	−104 (66)	−9 (92)	13 (93)	−164 (132)	−6 (48)
$t=5$	50 (83)	−244 (73)	−146 (70)	20 (96)	* (*)	−211 (132)	1 (48)
$t=6$	17 (90)	−286 (84)	−172 (79)	−3 (111)	* (*)	−189 (112)	−21 (48)
Ave. 1 to 6	38 (64)	−291 (54)	−166 (56)	−25 (83)	115 (78)	−175 (108)	16 (37)
				Adult females			
$t=1$	55 (36)	−69 (33)	−73 (29)	40 (30)	167 (35)	−84 (56)	26 (28)
$t=2$	55 (39)	−9 (33)	−15 (29)	63 (34)	122 (40)	−57 (69)	9 (36)
$t=3$	31 (52)	5 (34)	−6 (31)	42 (40)	98 (40)	−62 (70)	−13 (37)
$t=4$	35 (45)	5 (34)	−10 (34)	21 (47)	87 (43)	−42 (60)	2 (35)
$t=5$	48 (45)	14 (38)	−6 (37)	26 (48)	* (*)	−38 (63)	1 (31)
$t=6$	16 (42)	−10 (37)	−24 (37)	−2 (44)	* (*)	−35 (58)	−2 (34)
Ave. 1 to 6	40 (38)	−11 (31)	−22 (29)	32 (35)	119 (39)	−53 (57)	4 (30)
				Male youth			
$t=1$	8 (61)	−41 (56)	−40 (53)	37 (61)	302 (120)	−29 (106)	104 (44)
$t=2$	28 (55)	10 (62)	9 (63)	45 (65)	275 (140)	12 (110)	36 (43)
$t=3$	−8 (77)	−29 (74)	−24 (76)	10 (83)	217 (153)	38 (136)	70 (48)
$t=4$	4 (71)	2 (69)	8 (70)	30 (81)	157 (176)	110 (162)	116 (45)
$t=5$	42 (76)	63 (72)	73 (71)	46 (99)	* (*)	132 (182)	95 (48)
$t=6$	−31 (92)	9 (76)	21 (75)	−68 (131)	* (*)	−63 (210)	108 (53)
Ave. 1 to 6	7 (53)	2 (52)	8 (52)	17 (70)	238 (144)	33 (128)	88 (38)
				Female youth			
$t=1$	−8 (46)	3 (34)	17 (32)	60 (45)	−11 (72)	74 (76)	55 (32)
$t=2$	27 (49)	46 (39)	54 (39)	81 (46)	−31 (79)	91 (77)	52 (32)
$t=3$	49 (52)	64 (42)	72 (41)	101 (51)	−37 (82)	84 (90)	74 (34)
$t=4$	−28 (59)	18 (48)	18 (47)	48 (57)	−55 (85)	−33 (119)	21 (36)
$t=5$	8 (54)	46 (43)	48 (41)	46 (56)	* (*)	21 (131)	37 (39)
$t=6$	1 (62)	37 (50)	40 (48)	38 (62)	* (*)	−3 (114)	57 (36)
Ave. 1 to 6	8 (42)	36 (36)	41 (35)	62 (42)	−34 (78)	39 (83)	49 (26)

† A 2% trimming rule is used for adult males and females to determine the overlapping support region (see Appendix C) and a 5% trimming rule is used for male and female youth. A fixed bandwidth of 0·06 and a biweight kernel, described in Appendix A, are used to compute the estimates for all four groups. Bootstrapped standard errors are shown in parentheses. They are based on 50 replications with 100% sampling.
* Data not available to compute for this quarter. Averages reported over available quarters.
†† The regular predictor model is the model for the probability of programme participation that maximizes the percent correctly classified. The regressors in the model for each demographic group are given in the footnote to Table 2.
‡ Coarse I predictor model includes indicator variables for site, race, age, education, marital status, and for the presence of children aged less than 6 in the household. Coarse II predictor model augments Coarse I with earnings from the year preceding enrollment into the programme. Coarse III predictor model augments Coarse I with indicators for labour force transition patterns.
§ SIPP predictor model includes indicators for age, race, education, marital status, children aged less than 6, labour force transition patterns and levels of earnings in the preceding year. The data used are SIPP JTPA eligibles matched with Experimental JTPA Controls.
§§ Site Mismatch predictor model is the same as the regular predictor model. The data used are Controls from Providence and Jersey City matched with ENPs from Corpus Christi and Fort Wayne.
§§§ The data used are Experimental JTPA Controls matched with experimental JTPA Persons assigned to Treatment who enrolled in JTPA but dropped out before receiving services. No-show predictor model includes indicator variables for site and recommended training services. For adult males, the model also includes earnings last year, earnings squared, indicators for preferred language Spanish and for preferred language other than Spanish or English, and an indicator for whether enrollment in JTPA was required. For adult women, the model also includes indicators for last employed 0–6 months ago and for 7–12 months ago and an indicator for whether enrollment in JTPA was required. For male youth, the model also includes indicator variables for race and for the quarter of the year. For female youth, the model also includes an indicator for race. (Models were chosen to maximize the percent correctly classified using available variables—see Appendix E available on request from the authors).

TABLE 6(b)

Bias from difference-in-differences local linear regression estimator†
Under alternative predictor models for the probability of programme participation

Quarter	Regular‡	Coarse I‡	Coarse II‡	Coarse III†	SIPP‡	Site mismatch‡
			Adult males			
$t=1$	104 (63)	167 (67)	31 (57)	67 (68)	−97 (38)	−135 (126)
$t=2$	77 (92)	143 (82)	−80 (62)	103 (107)	−230 (51)	−72 (130)
$t=3$	74 (114)	62 (95)	−158 (71)	105 (134)	−277 (52)	−9 (141)
$t=4$	98 (91)	33 (93)	−150 (82)	47 (109)	−338 (72)	19 (151)
$t=5$	−5 (99)	−73 (104)	−254 (86)	−29 (122)	* (*)	−136 (167)
$t=6$	−35 (111)	−143 (106)	−255 (96)	−36 (129)	* (*)	−82 (165)
Ave. 1 to 6	52 (74)	32 (78)	−144 (61)	43 (95)	−236 (45)	−69 (123)
			Adult females			
$t=1$	74 (30)	80 (24)	71 (23)	86 (25)	−43 (10)	38 (42)
$t=2$	60 (39)	69 (35)	54 (33)	85 (35)	−86 (25)	66 (56)
$t=3$	14 (59)	20 (39)	−1 (37)	25 (49)	−99 (27)	43 (66)
$t=4$	7 (56)	−16 (42)	−34 (40)	−15 (59)	−116 (36)	67 (62)
$t=5$	23 (53)	−9 (42)	−26 (42)	−5 (55)	* (*)	80 (68)
$t=6$	−18 (51)	−44 (42)	−52 (43)	−40 (53)	* (*)	48 (72)
Ave. 1 to 6	27 (39)	17 (31)	2 (30)	23 (39)	−86 (21)	57 (50)
			Male youth			
$t=1$	80 (77)	123 (56)	111 (56)	194 (85)	22 (53)	−92 (98)
$t=2$	61 (60)	102 (73)	81 (72)	58 (72)	12 (78)	1 (118)
$t=3$	70 (86)	−9 (88)	−23 (87)	38 (100)	−60 (102)	33 (152)
$t=4$	−5 (77)	−45 (81)	−54 (80)	−6 (85)	−85 (135)	32 (157)
$t=5$	−11 (81)	34 (85)	28 (82)	−3 (108)	* (*)	25 (188)
$t=6$	−64 (84)	18 (83)	19 (80)	−74 (126)	* (*)	−117 (211)
Ave. 1 to 6	22 (48)	37 (56)	27 (54)	34 (59)	−28 (81)	−20 (122)
			Female youth			
$t=1$	−14 (41)	59 (39)	62 (41)	14 (36)	−14 (31)	5 (66)
$t=2$	27 (47)	82 (42)	75 (42)	48 (47)	−67 (33)	29 (88)
$t=3$	83 (58)	116 (51)	106 (52)	91 (62)	−90 (46)	89 (111)
$t=4$	4 (59)	53 (48)	36 (48)	30 (56)	−96 (60)	−21 (139)
$t=5$	−7 (63)	−1 (43)	−8 (43)	−16 (60)	* (*)	44 (154)
$t=6$	6 (69)	2 (53)	5 (53)	−3 (71)	* (*)	2 (134)
Ave. 1 to 6	17 (39)	52 (35)	46 (35)	28 (39)	−67 (35)	25 (87)

† A 2% trimming rule is used to determine the overlapping support region for adult groups and a 5% trimming rule is used for the youth groups (see Appendix C). A fixed bandwidth equal to 0·06 and a biweight kernel, described in Appendix A, are used to compute the nonparametric estimates.
* Data not available to compute for these periods. Averages are reported over available quarters.
‡ The alternative predictor models for the probability of programme participation are described in the footnote to Table 6(a).

III model for $P(X)$ and of comparable magnitude for the other models and for female youth the richest model for $P(X)$ has the lowest estimated bias.

Table 6(b) presents analogous bias estimates for the conditional difference-in-differences estimator. For the richest conditioning information ("regular" $P(X)$ models) the difference-in-differences estimator sometimes exhibits lower bias, but for other conditioning sets the bias varies. For this estimator, the choice of X in $P(X)$ makes less of a difference to the effectiveness of the method in our samples and is an attractive feature of it.

We have performed a comparable sensitivity analysis for variations in the regressors included in the outcome equations (T), including no regressors. These results are available on request from the authors. We find little variation in estimated biases across alternative specifications of the outcome model.

14. THE IMPORTANCE OF GEOGRAPHIC MISMATCH AND NONUNIFORMITY OF THE SURVEY INSTRUMENT

We now examine the empirical importance of two additional sources of bias that plague nonexperimental evaluations: mismatch of the geographic location of programme participants and comparison group members and mismatch of the survey instrument used to collect participant and comparison group data. Participants, experimental controls and nonexperimental comparison group members were administered the same questionnaire and reside in the same labour markets. Therefore they are not subject to these sources of bias. To investigate this problem we consider alternative comparison group samples drawn from SIPP (Survey of Income and Programme Participation).

The SIPP data are far richer than those used to construct nonexperimental comparison groups in previous evaluations of job training programmes. It is longitudinal with monthly observations on both earnings and employment and the samples are large. It also contains enough information to determine eligibility status for JTPA, which requires very detailed information about family structure, six-months earnings histories and welfare participation (see Devine and Heckman (1996)). As noted in Table 1, few major evaluations of job training programmes have had rich enough data to determine the eligibility of comparison group members. We restrict our SIPP comparison group samples to persons who meet the eligibility requirements of the JTPA programme. Like the ENP comparison group samples, the SIPP samples consist of eligible nonparticipating persons.

14.1. *The SIPP data*

We use the 1988 SIPP panel data set with observations that span from October, 1987 to December, 1989. The 1988 panel covers the time period of the JTPA experiment where random assignment took place from November, 1987 to September, 1989. The data contain earnings and labour force histories and information on participation in JTPA, although few JTPA participants are found in the data in any month. Appendix E, available on request, describes the SIPP data in detail.

A drawback in using SIPP as a comparison group for the JTPA data is that it is a broad, nationally-representative sample while the JTPA experiment was conducted in a few cities of moderate size. To protect confidentiality, information about the exact location of respondents was suppressed so it is not possible to find SIPP comparison group members in the same labour market as JTPA programme participants. Smith (1995) compares the earnings of the SIPP and ENP comparison groups. He finds that mean differences in earnings levels remain even after a variety of geographic and local labour market matching schemes are tried and that these differences cannot be explained solely by differences in local labour market variables like local unemployment rates.

We compare the earnings of JTPA experimental controls with those of the JTPA-eligible SIPPs. They are substantially different even after matching on region of residence or local unemployment rates in region of residence. However employment rates and labour force participation rates are similar between the two groups, suggesting that the observed earnings differences are due to cost of living differences or to differences in the survey questions on earnings. As shown in the middle panel of Table 2, with the exception of adult males, the bias in the raw data (B) is greater using the SIPP samples compared to using the ENP samples. For adults, the component of bias B attributable to selection B_3, is both absolutely and proportionately larger in the SIPP-control data than in the ENP-control data. The uncontrolled heterogeneity in questionnaires and locations across ENP